

UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

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PROJECT OBJECTIVES

- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input

COMPONENT SPECIFICATIONS

(x7) ESP32-WROVER-E

[datasheet](https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e_esp32-wrover-ie_datasheet_en.pdf)

- 2 data processors, 3 audio processors, 1 oscilloscope processor, 1 audio output/master coordinator.
- 3.3V
- 8MB PSRAM, 8MB Flash

(x11) VL53L4CX Time-of-Flight Sensor

[listing](<https://www.adafruit.com/product/5425>)

- 3.3V
- I2C communication
- SCL, SDA, GPIO, XSHUT pins

(x1) MAX9814 microphone

[documentation](<https://cdn-shop.adafruit.com/datasheets/MAX9814.pdf>)

(x2) MAX98357 Amplifier

[documentation](<https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp>)

(x1) 128x64 STEMMA QT OLED display

[documentation](<https://learn.adafruit.com/monochrome-oled-breakouts>)

PCM1794A DAC

[datasheet](https://www.ti.com/lit/ds/symlink/pcm1794a.pdf?ts=1739053305998&ref_url=https%253A%252F%252Fhr.mouser.com%252F)

OPA1612 I/V CONVERTER

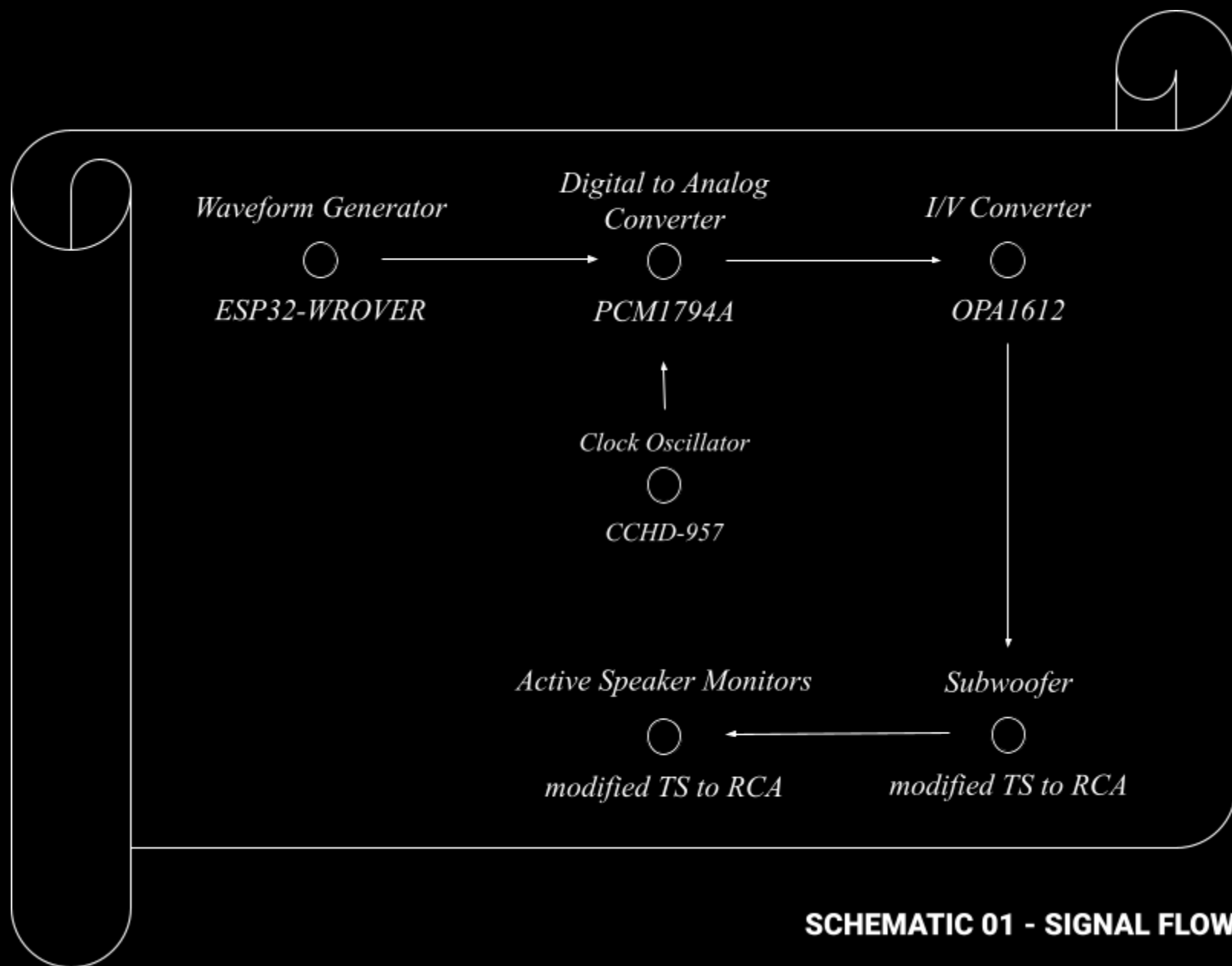
[datasheet](https://www.ti.com/lit/ds/symmlink/opa1612.pdf?ts=1739154487554&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FOPA1612)

CURRENT OBJECTIVES

01. SCHEMATIC: ESP32 waveform generator > DAC (Digital to Analog Converter) > I/V Converter > TS/RCA > Subwoofer > RCA/TS > monitor speakers

- Create custom symbols for:
 - DAC (PCM1794A)
 - I/V Converter (Burr Brown/OPA1612)
 - External Clock Oscillator (Crystek CCHD-957)

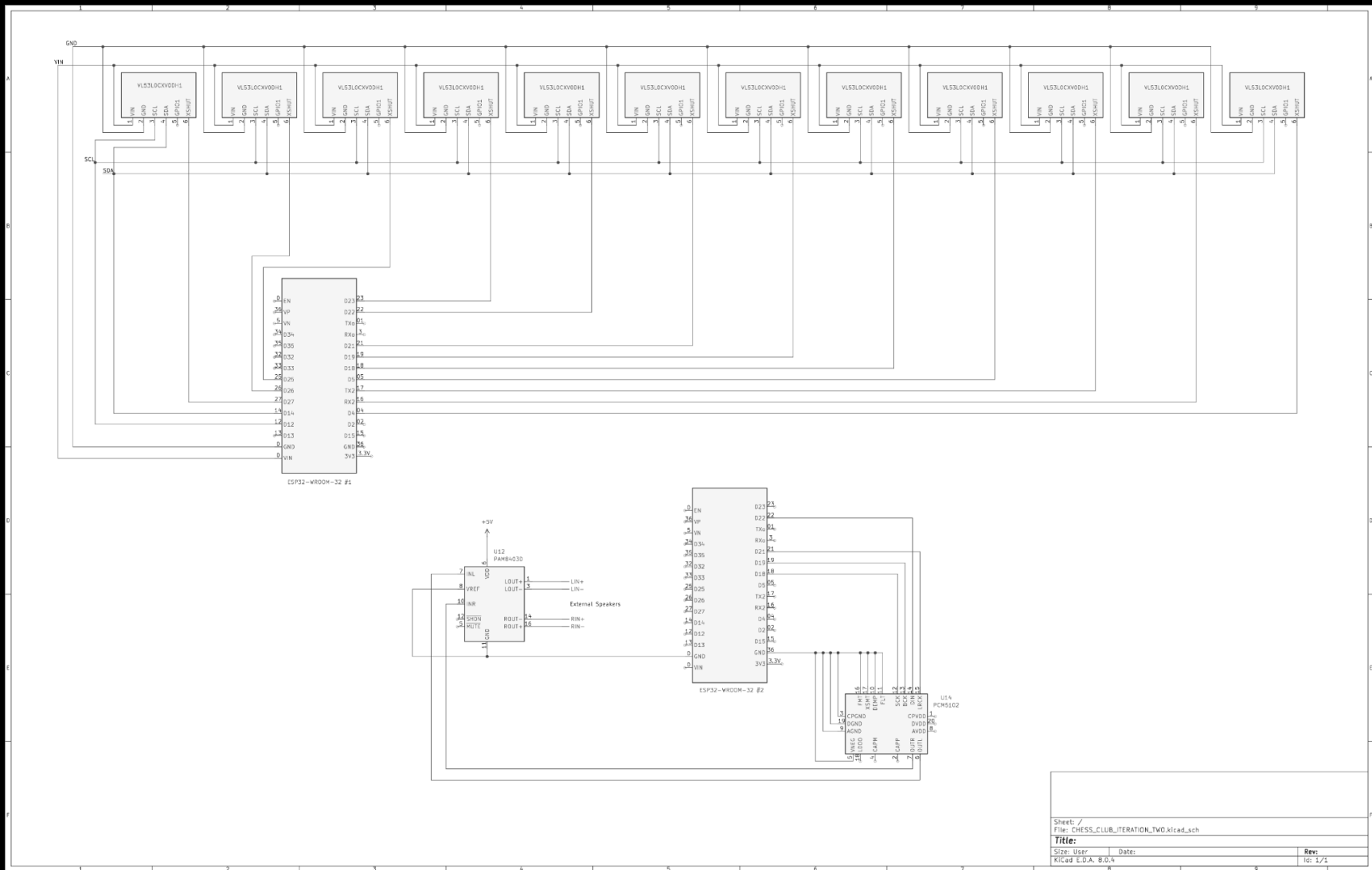
02. SCHEMATIC: Powering systems - Monitors, sub, ESP32s



SCHEMATIC 01 - SIGNAL FLOW

ESP32 WROVER > DAC > I/V Converter > Subwoofer > Active Monitor Speakers
Clock Oscillator > DAC

ARCHIVE



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

Preliminary Schematic Overview:

12 VL53LOCX proximity sensors connected to one ESP32.

PCM5102 DAC > ESP32 > PAM84030 Audio Driver > R/L External Speakers.

Note for modifications in the next version:

2 dedicated ESP32s for processing sensor data

VL53L4CX substitute

PRELIMINARY SCHEMATIC OVERVIEW

** power*

- 3.3V power supply
 - VCC + GND
- 20 power (VDD) terminals to 3.3V
 - 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speaker

** inputs*

- *sensors*
 - VIN to 3.3V
 - GND to GND
 - SCL to ESP32 SCL
 - SDA to ESP32 SDA
 - GPIO (none)
 - XSHUT to unique GPIO on ESP32
- *microphone*

** processors*

- ESP32s

- *amplifiers*
- *DAC (Digital-to-Analog Converter)*

*** *outputs***

- *speakers*
- *subwoofer*
- *oled display*